

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/Correction_X/C_DEB_SAAWYe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3330, C2 = 6666, C3 = 2212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,          0.2, 0.0001,          0.8);
Distrib (pAm,         TruncNormal_cv,        100,          0.2, 40,          2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,          0.1, 0.1,          2);
Distrib (v,           TruncNormal_cv,        0.18505,        0.1, 0.001,          2);
Distrib (kap,         Uniform,                0,            1);
Distrib (kapA,        Uniform,                0.1,          0.9);
Distrib (Ehp,         TruncNormal_cv,        100,          0.2, 40,          300);
Distrib (E_coc,       TruncNormal_cv,        30,          0.2, 1,          300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,        0.3);
Distrib (dv,          TruncNormal_cv,        2,          0.5, 0,          10);
Distrib (wE,          LogUniform,          0.000000001,    0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter, seeds)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kap.1.	kapA.1.
Result_mode	0.1123300	298.82700	0.28457200	0.18434900	0.92465000	0.47283800

2.5%	0.0987924	270.90255	0.20371595	0.17068200	0.88048100	0.34715000
97.5%	0.4903854	319.81700	0.35499400	0.23718800	0.93119515	0.57161835
Min	0.0717510	90.58550	0.15511000	0.13470000	0.01107090	0.15520200
Max	0.7922290	341.60300	0.87744000	0.26873900	0.94266300	0.64106500
Result_mean	0.2232302	295.25285	0.27801571	0.20340779	0.91111031	0.46235941
Result_sd	0.1053736	12.67551	0.03795232	0.01686738	0.01429167	0.05407865
	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	dv.1.	wE.1.	
Result_mode	41.541900	13.170800	0.0086825200	7.4993000	1.374080e-09	
2.5%	40.259800	9.893759	0.0022123600	6.1423800	1.088270e-09	
97.5%	65.250200	16.740000	0.0096913100	8.7795600	6.303620e-07	
Min	40.001300	8.238260	0.0000100268	0.5220400	1.020800e-09	
Max	107.006000	32.418600	0.0136630000	9.9531500	2.441410e-06	
Result_mean	47.773197	13.092012	0.0054304244	7.5809580	9.169919e-08	
Result_sd	6.734773	1.725483	0.0020077593	0.6853002	1.800469e-07	
	Sigma_W.1.					
Result_mode	0.17396400					
2.5%	0.15190300					
97.5%	0.19587490					
Min	0.14046300					
Max	5.67894000					
Result_mean	0.17298988					
Result_sd	0.04740581					

Number of observations, n = 277
 Number of parameters, k = 11
 Log-likelihood, LL = -79.01795
 BIC = 219.9001

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.03	1.09
E_coc.1.	1.14	1.41
Ehp.1.	1.11	1.33
Fm.1.	1.01	1.02
kap.1.	1.07	1.19
kapA.1.	1.21	1.61
pAm.1.	1.11	1.33
r_pAm_pM.1.	1.19	1.57
rOM_ClxHorse.1.	1.02	1.06
Sigma_W.1.	1.00	1.01

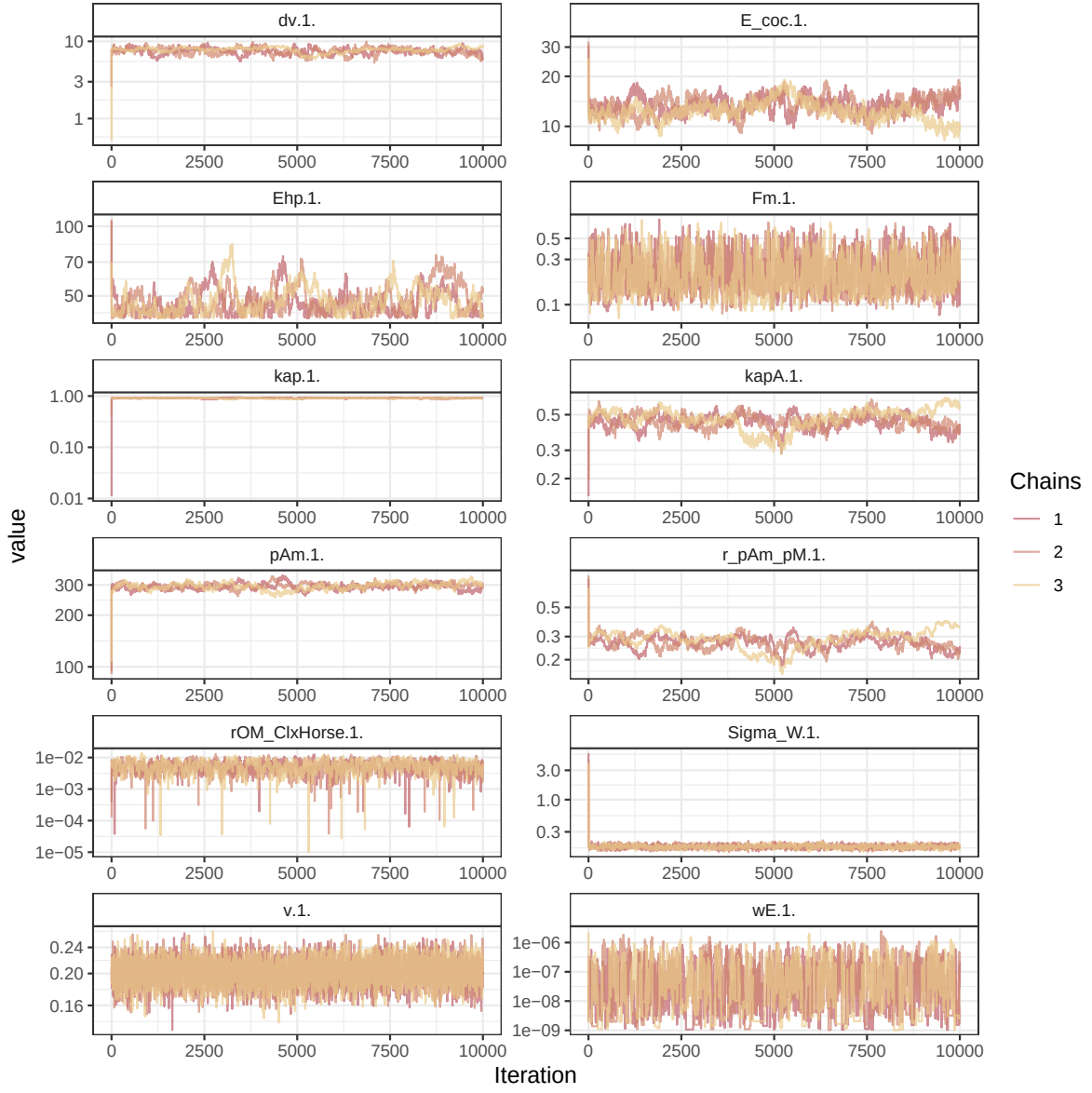


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

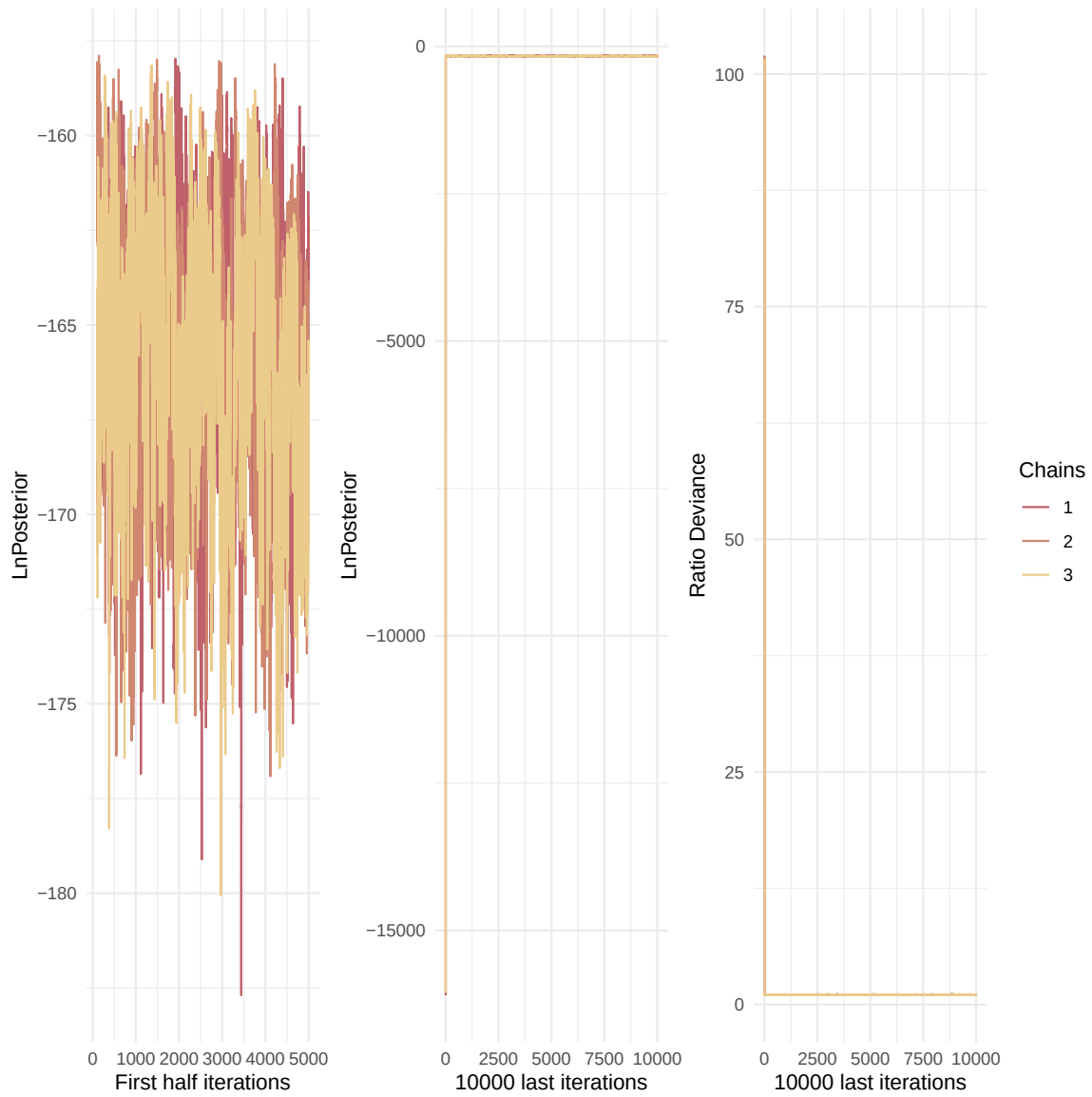


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

v.1.	1.01	1.02
wE.1.	1.03	1.06

Multivariate psrf

1.17

```
mcmc_list_corr <- mcmc.list(  
  lapply(mcmc_list, function(x) {  
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc  
  })  
)  
  
colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))  
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

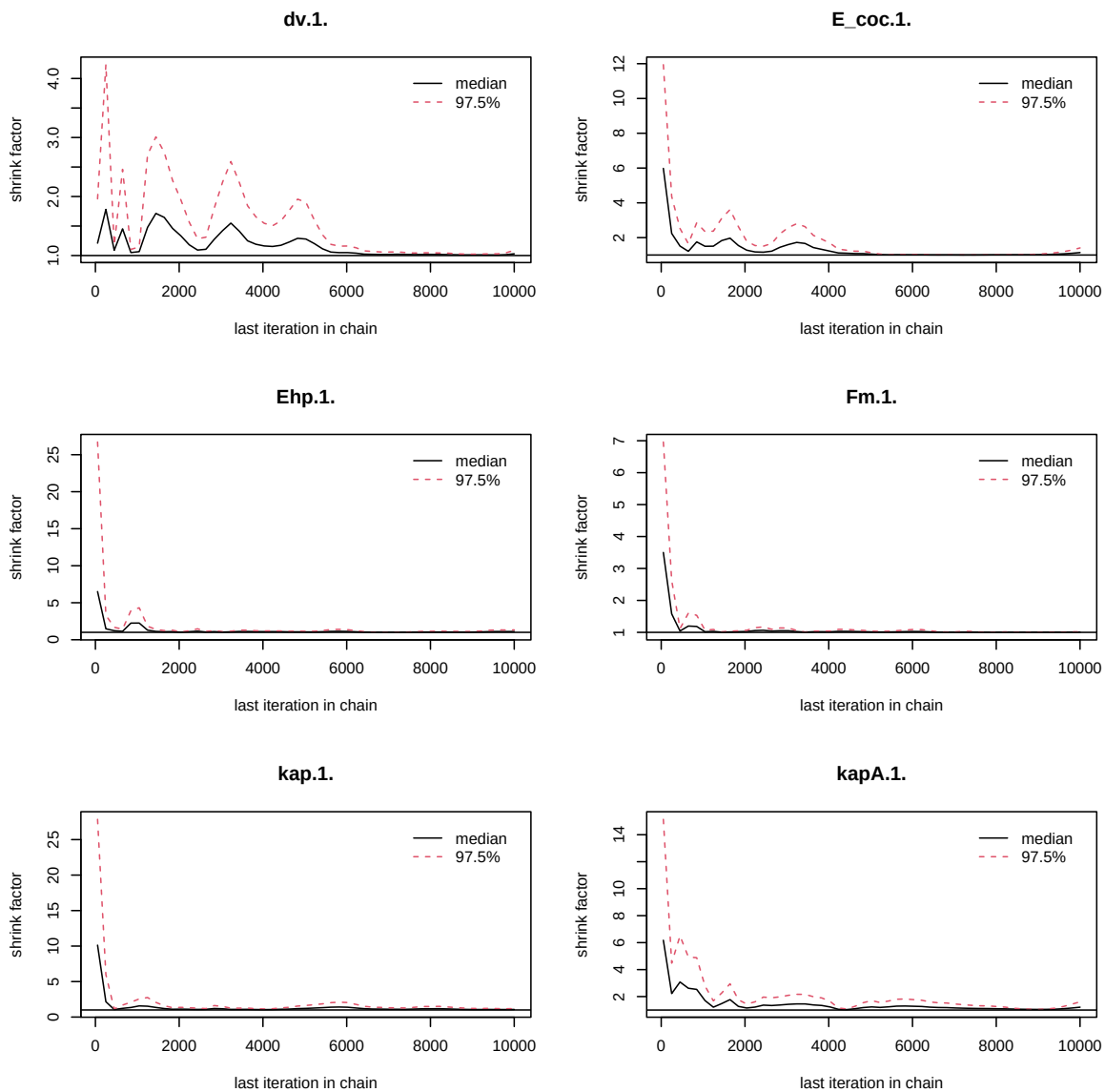


Figure 3: Chains convergence.

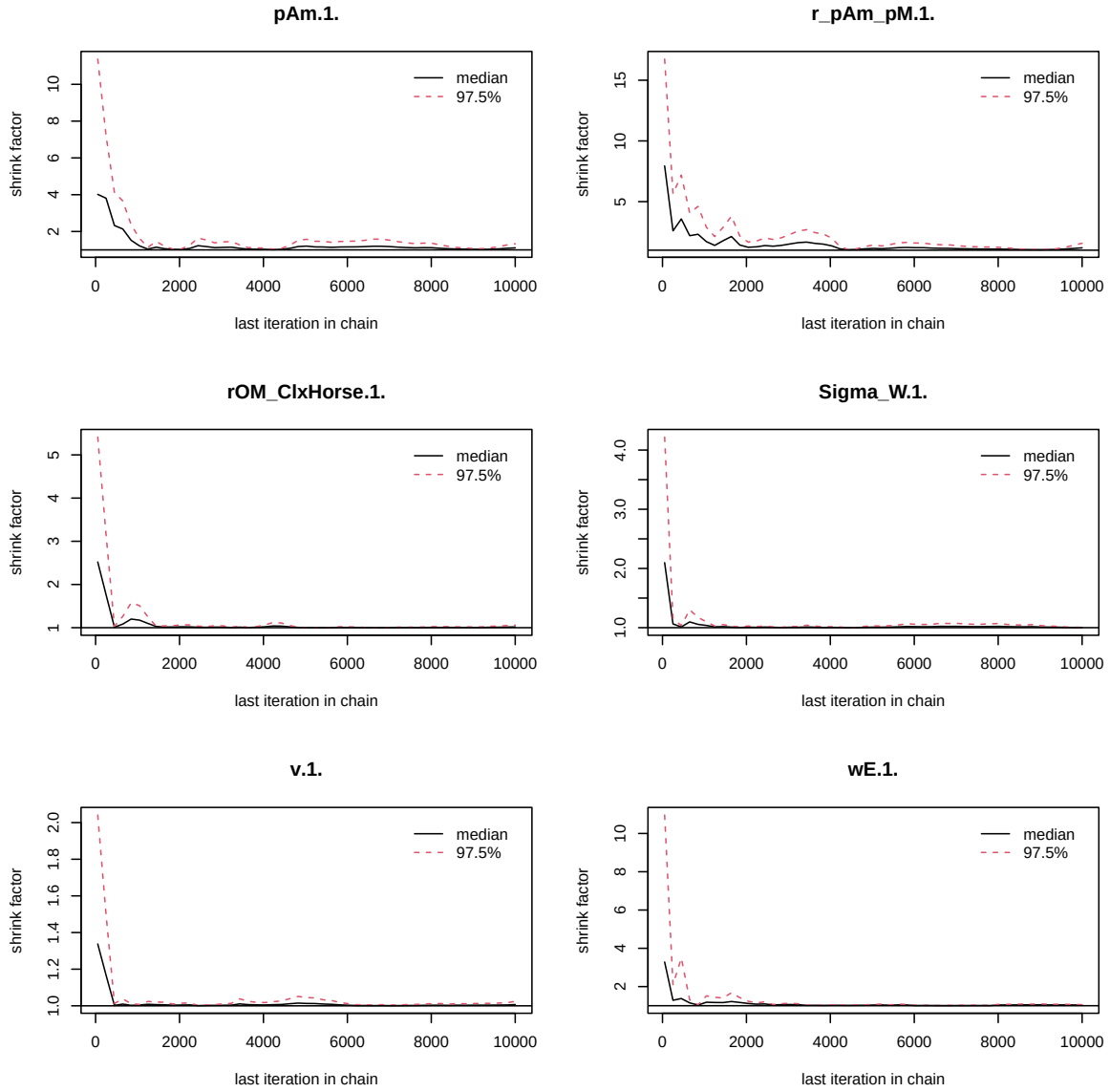


Figure 4: Chains convergence.

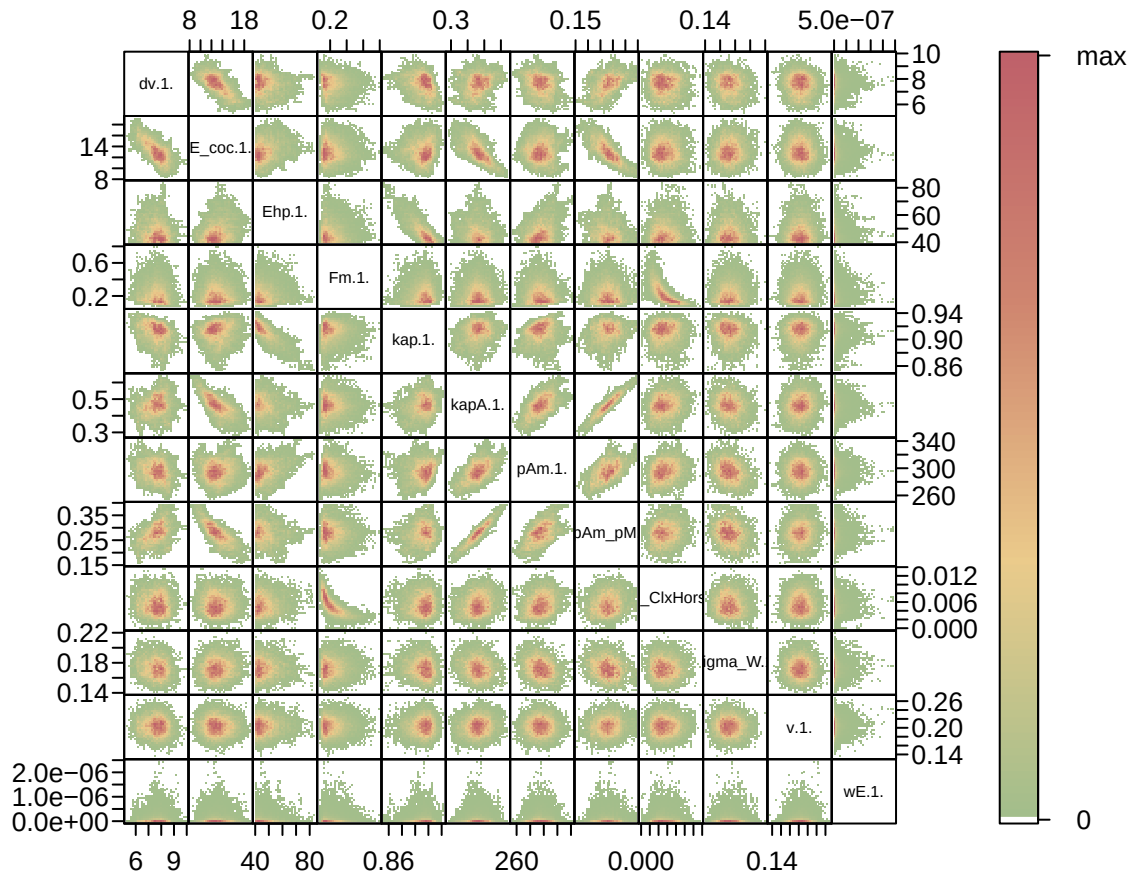


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

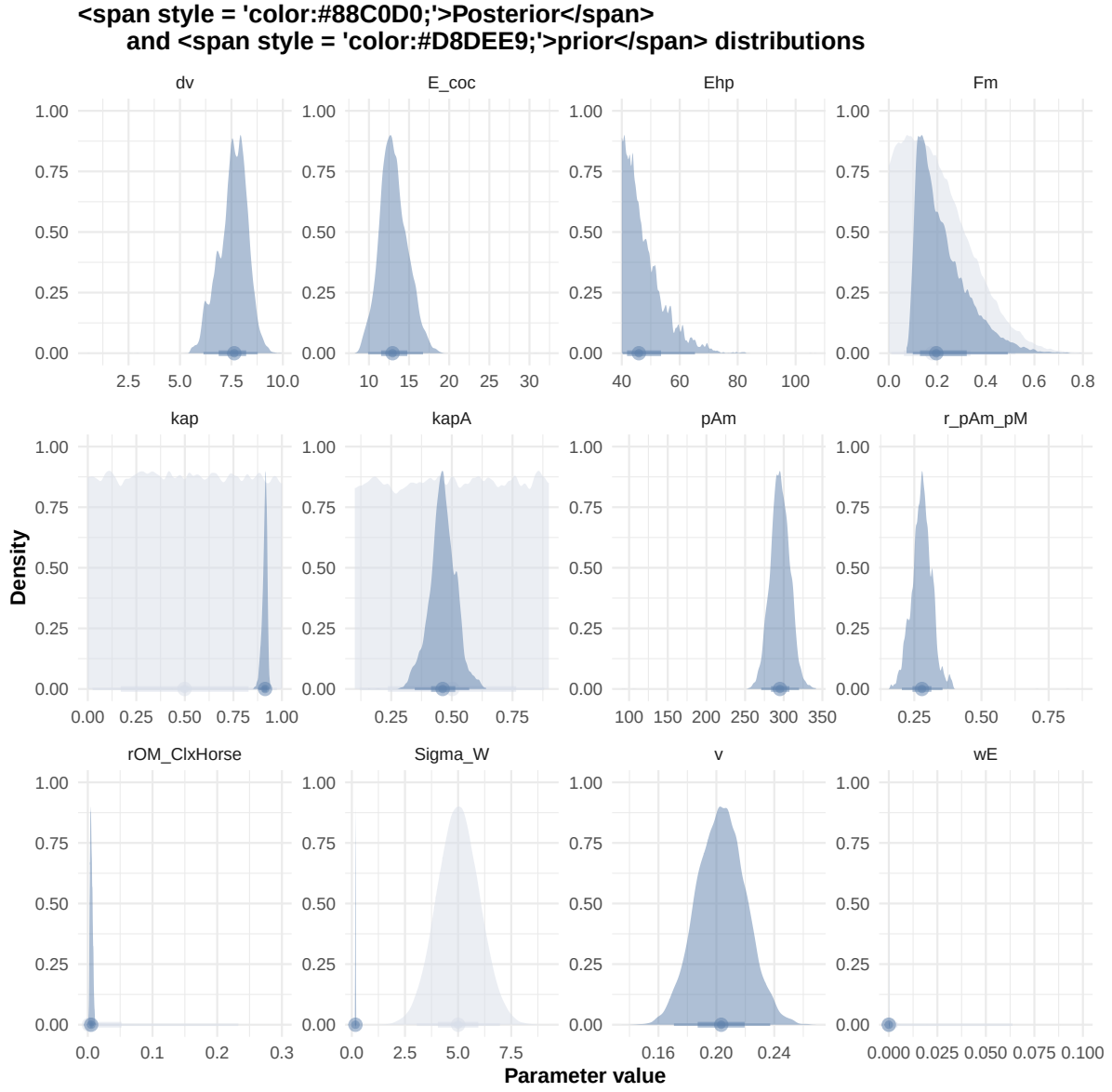


Figure 6: Distributions of the priors and the posteriors.


```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

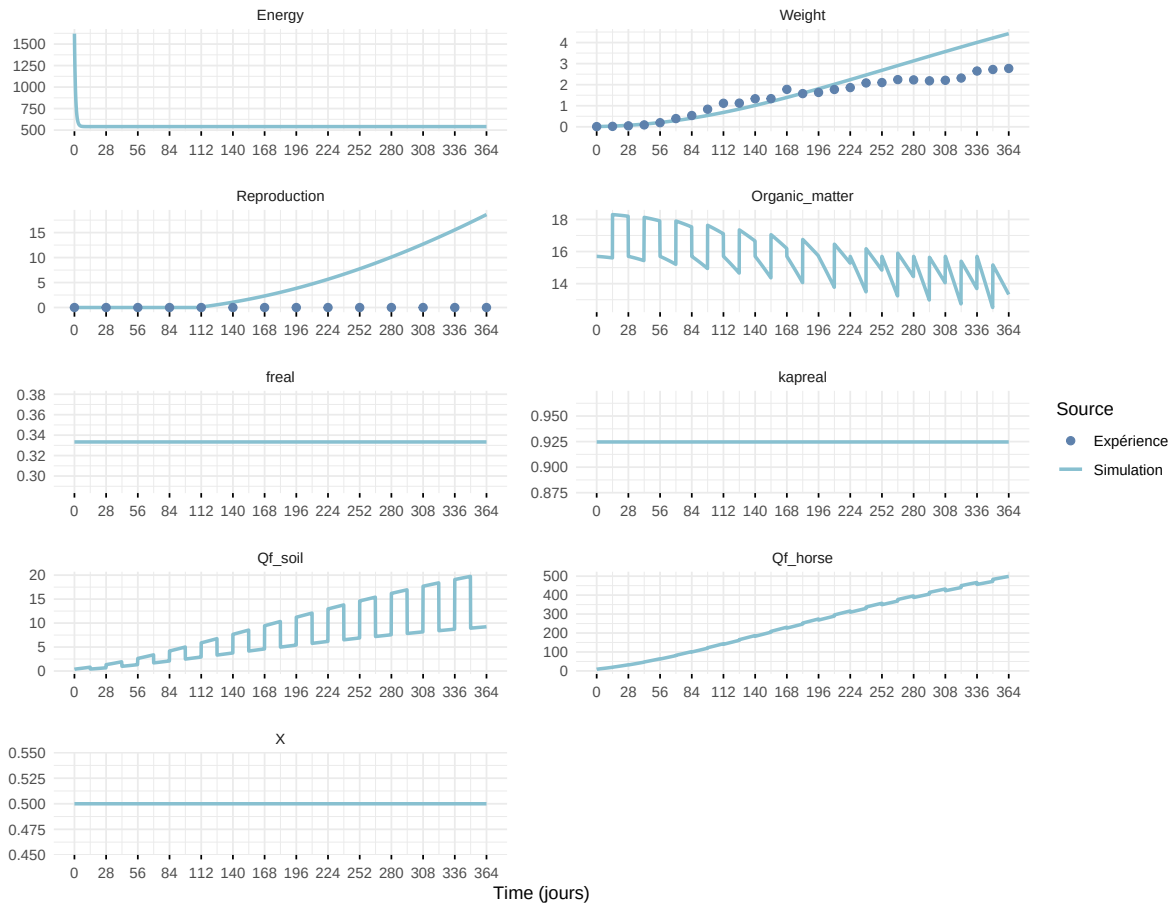
df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected join
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship =

"many-to-many" to silence this warning.

Expérience : Bart2019_XP1_1



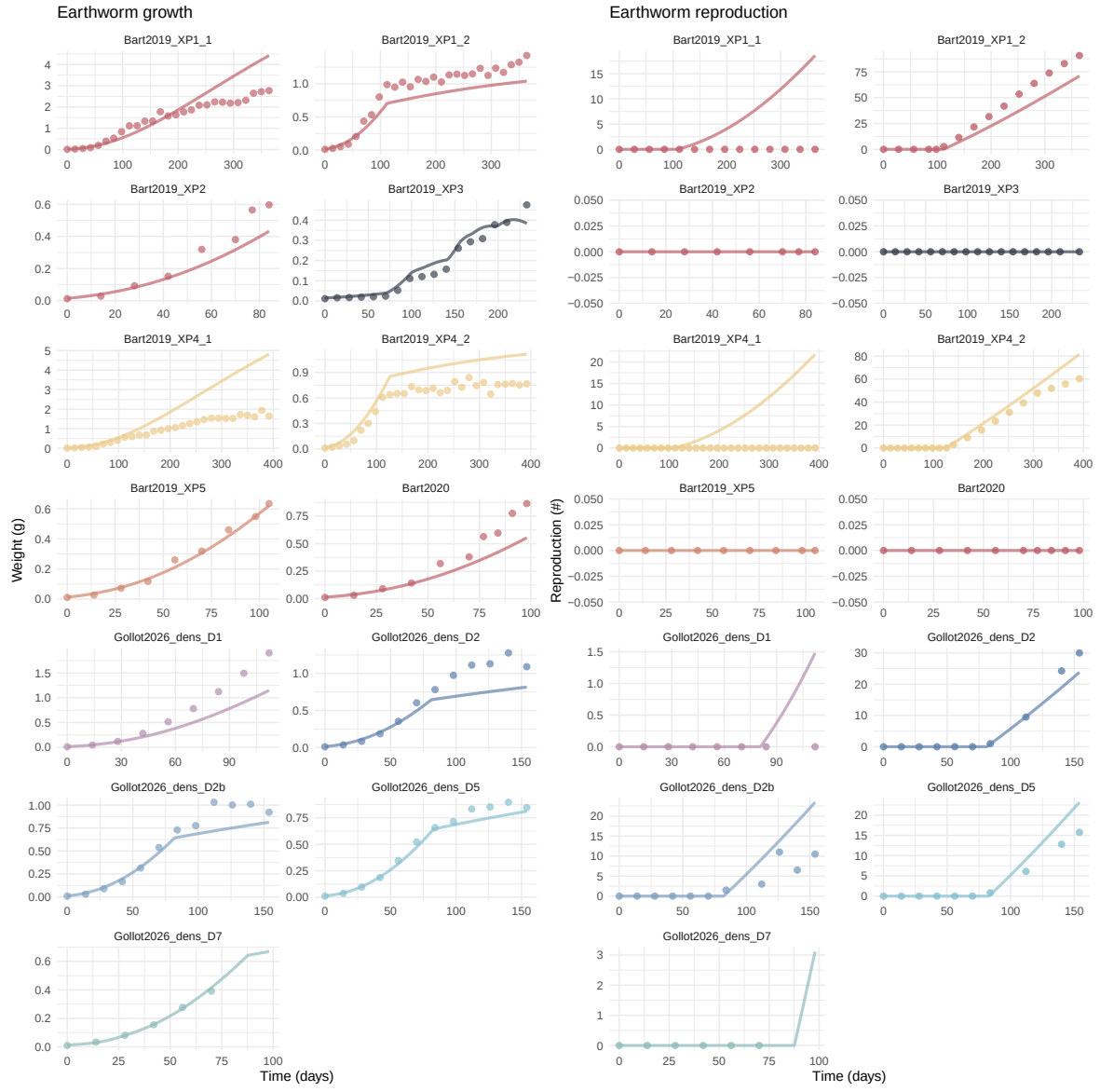
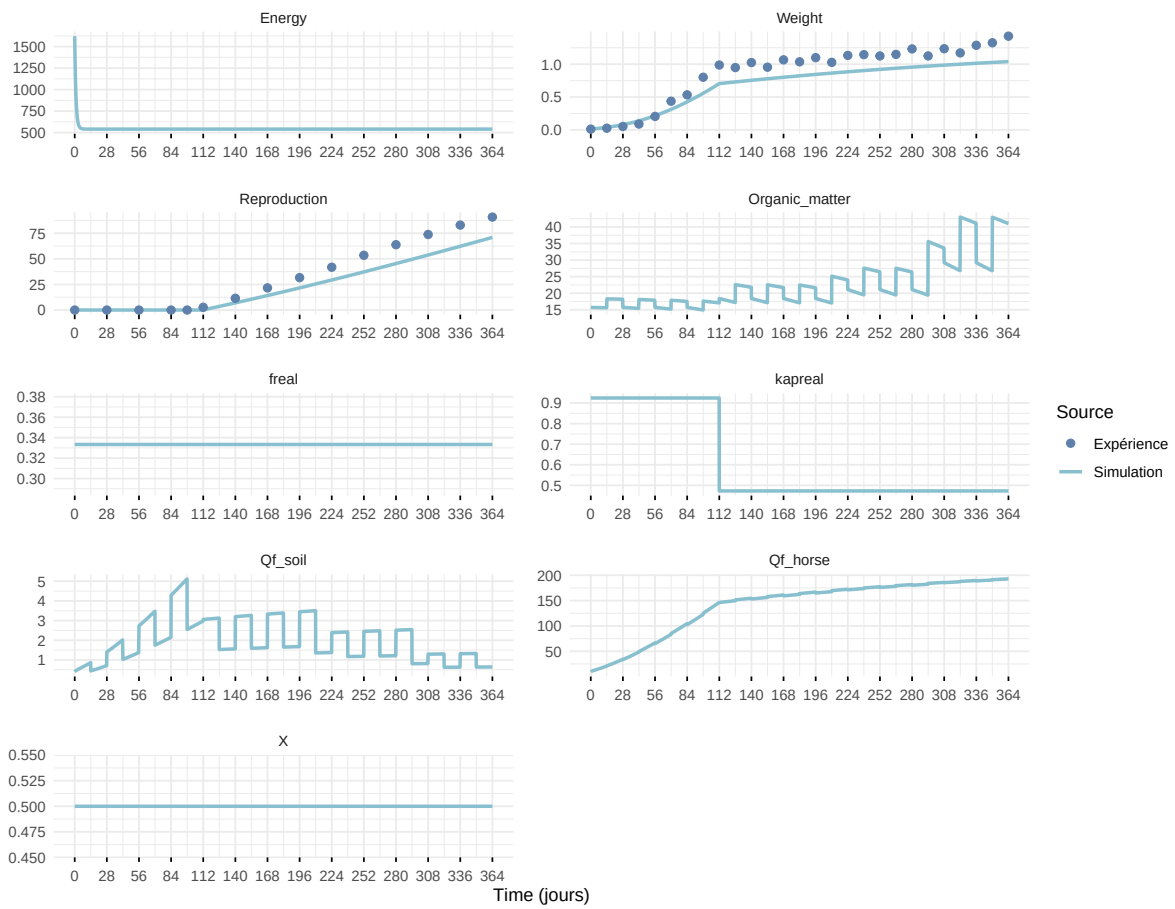
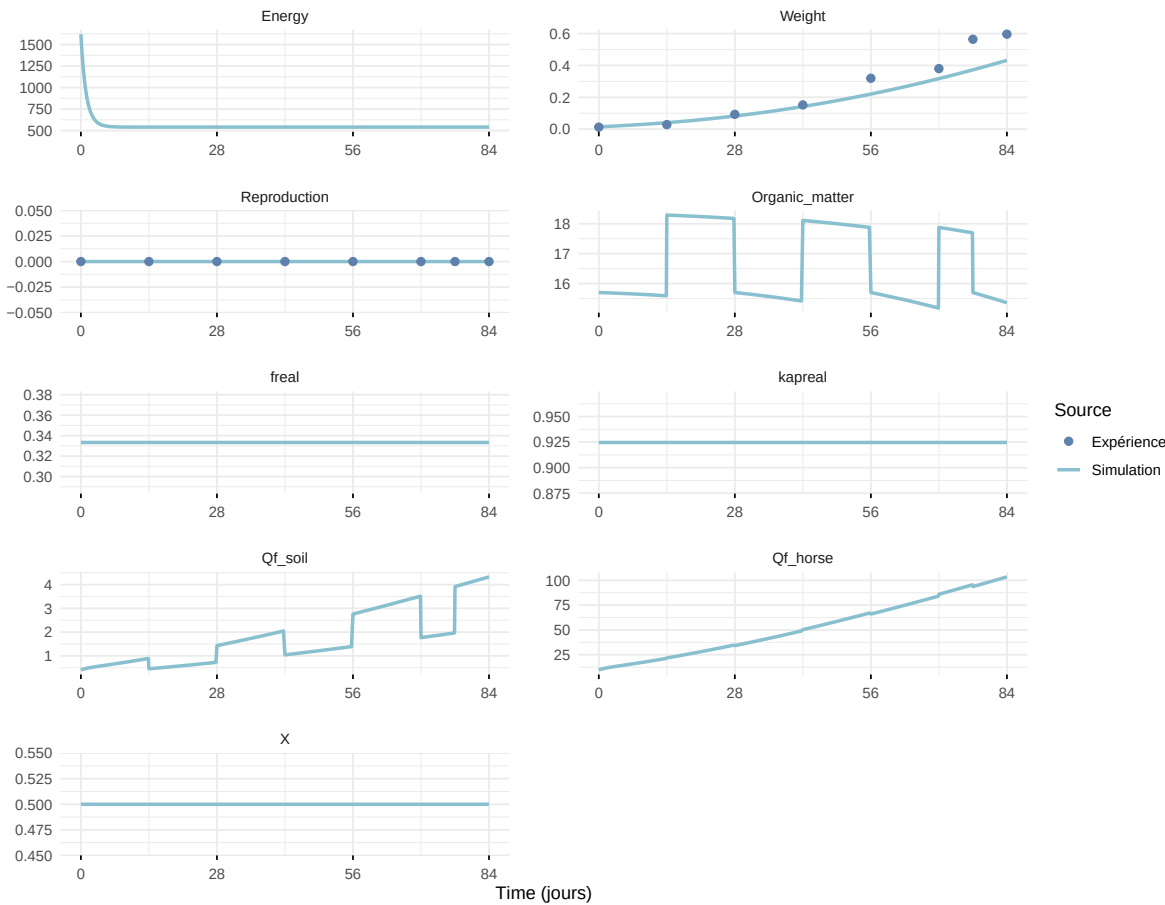


Figure 7: Simulations Weight and Reproduction.

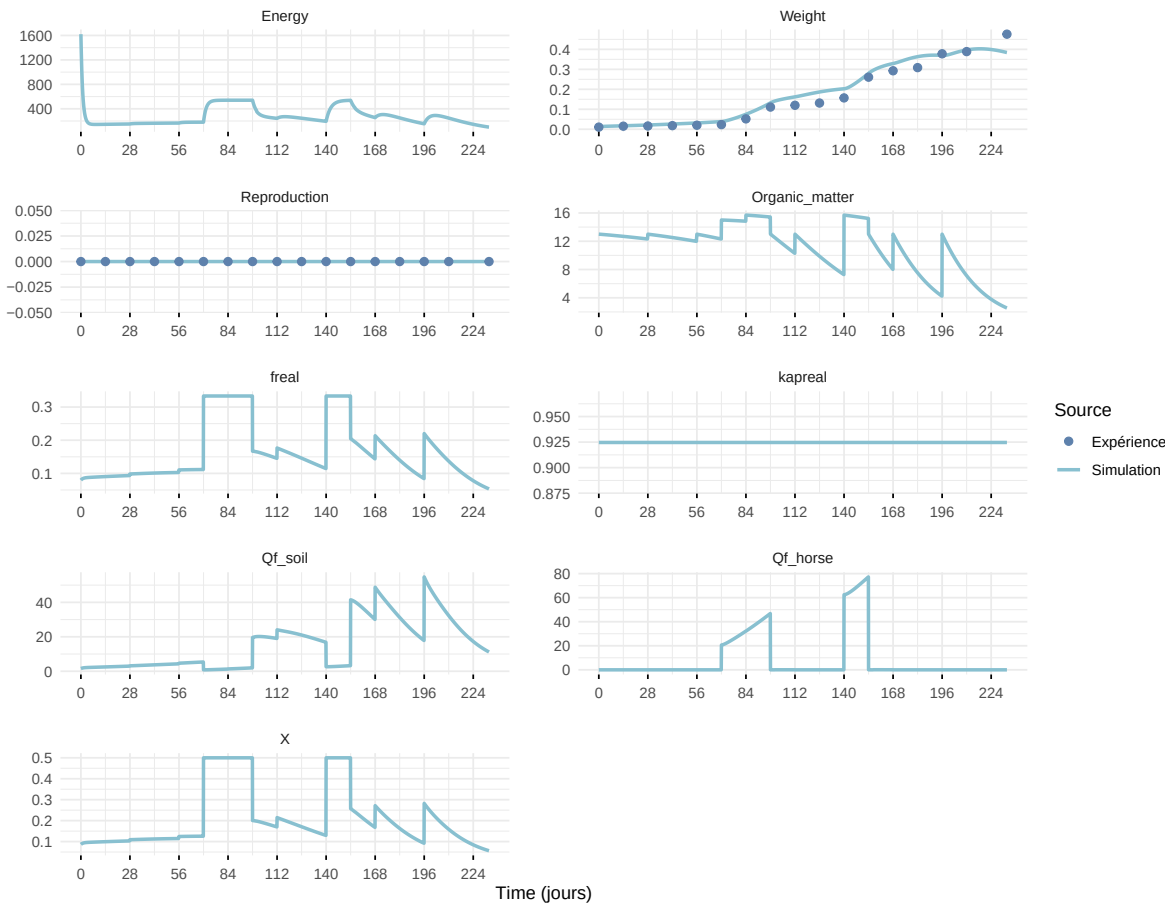
Expérience : Bart2019_XP1_2



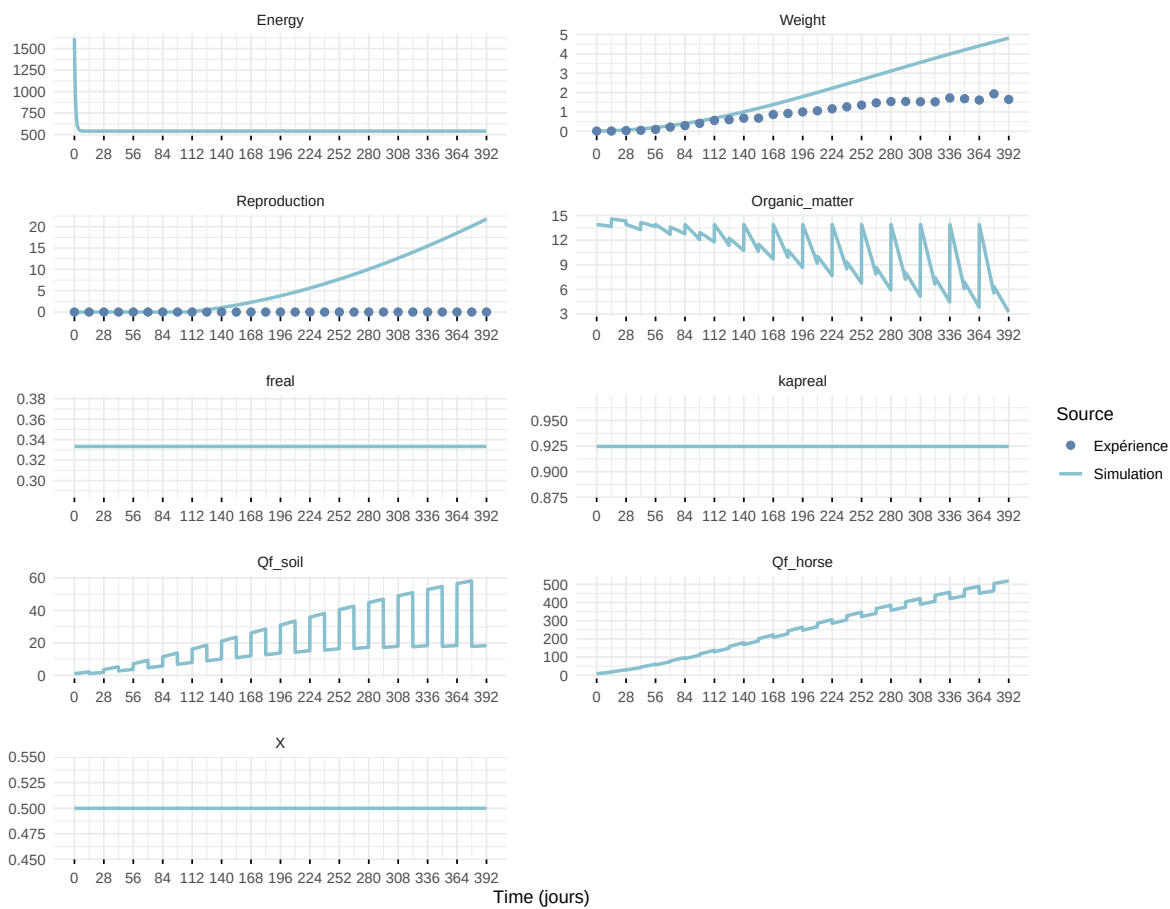
Expérience : Bart2019_XP2



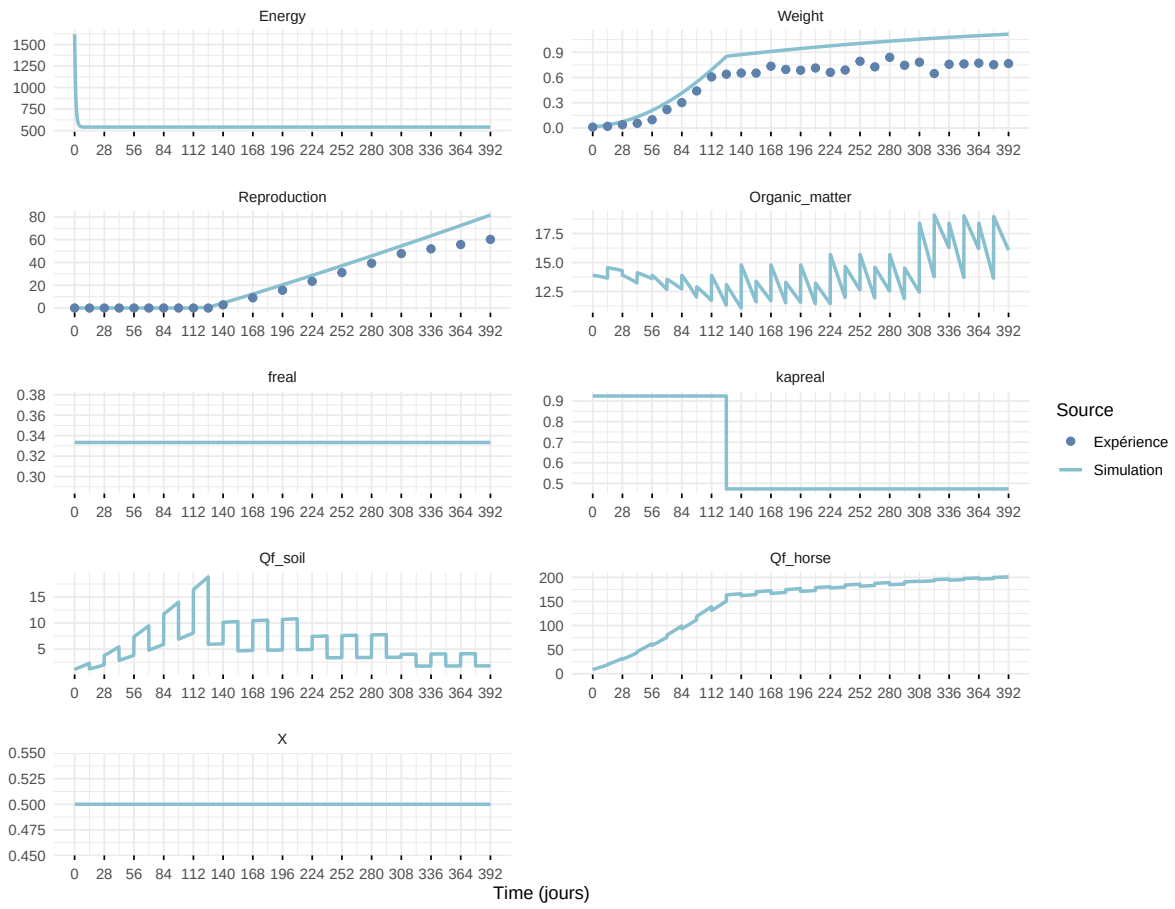
Expérience : Bart2019_XP3



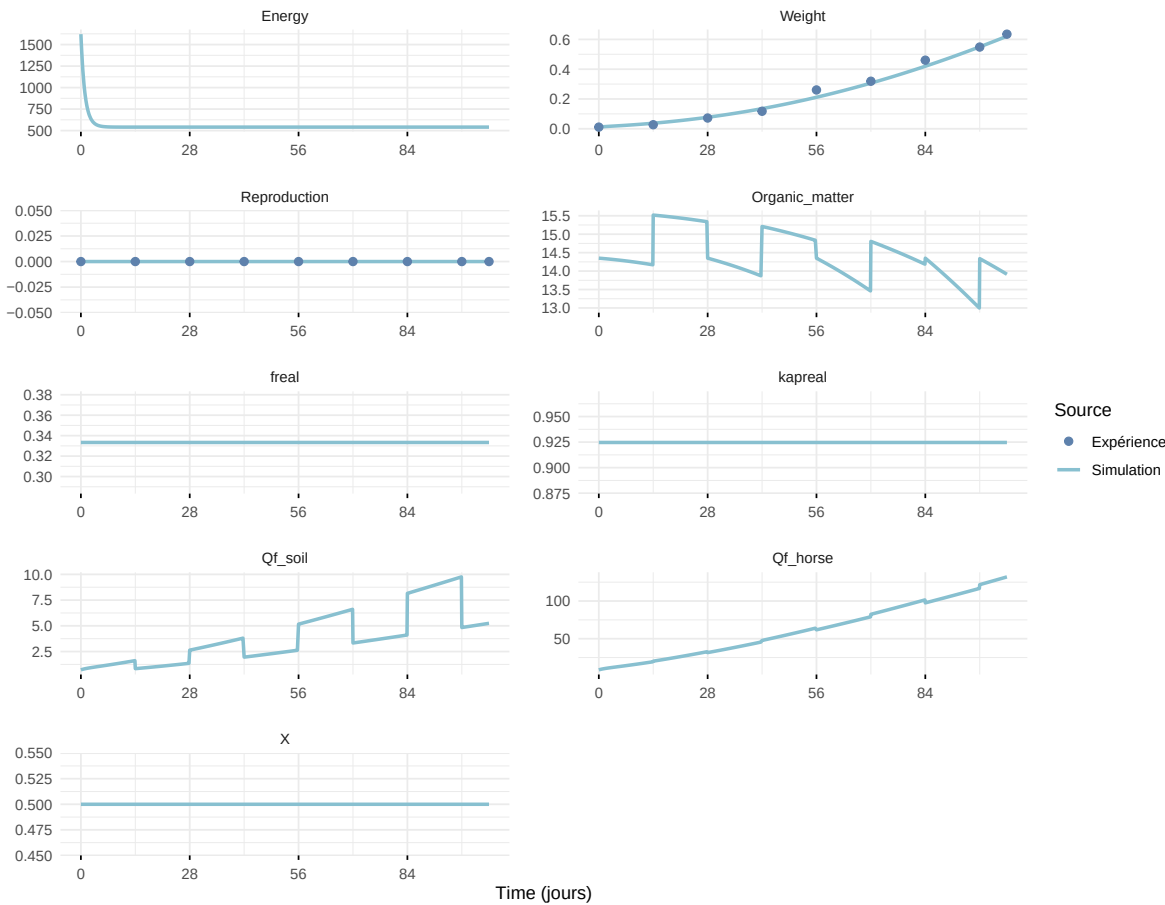
Expérience : Bart2019_XP4_1



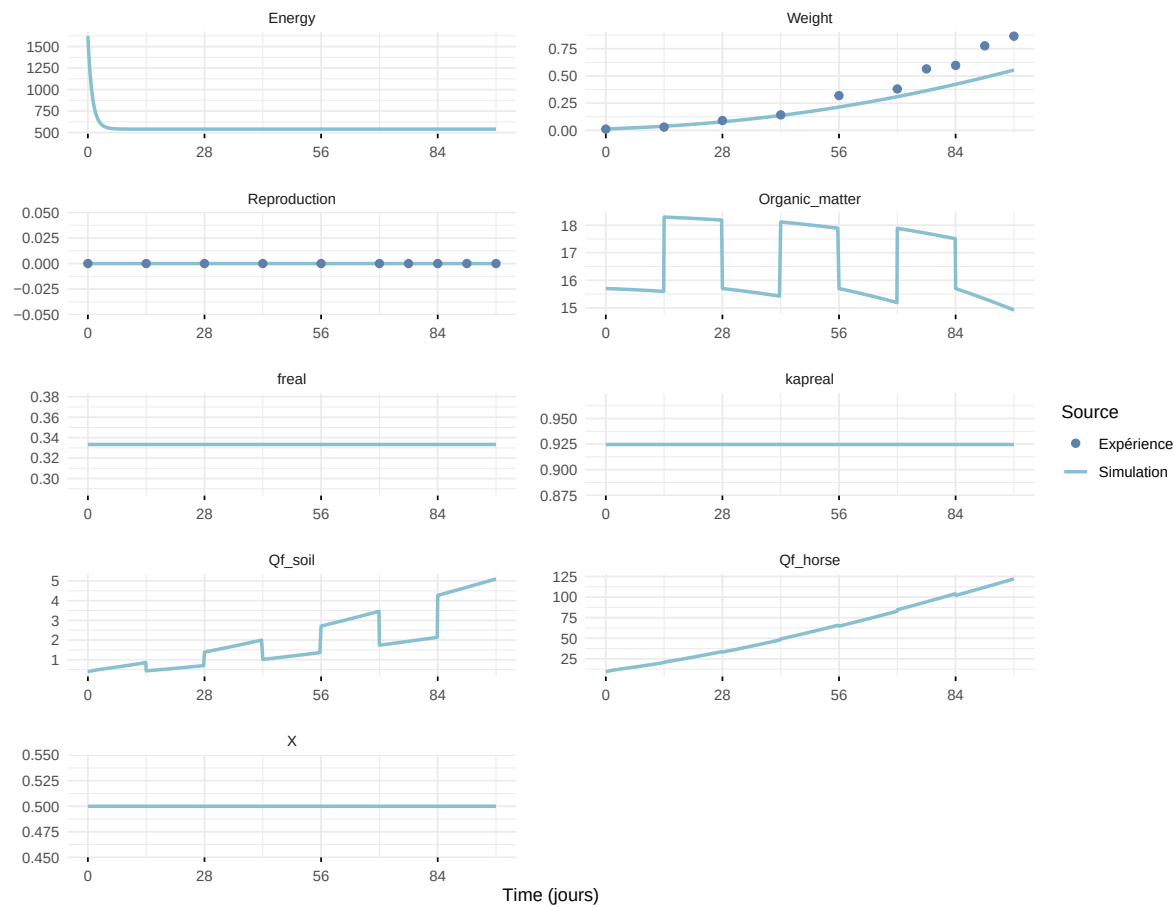
Expérience : Bart2019_XP4_2



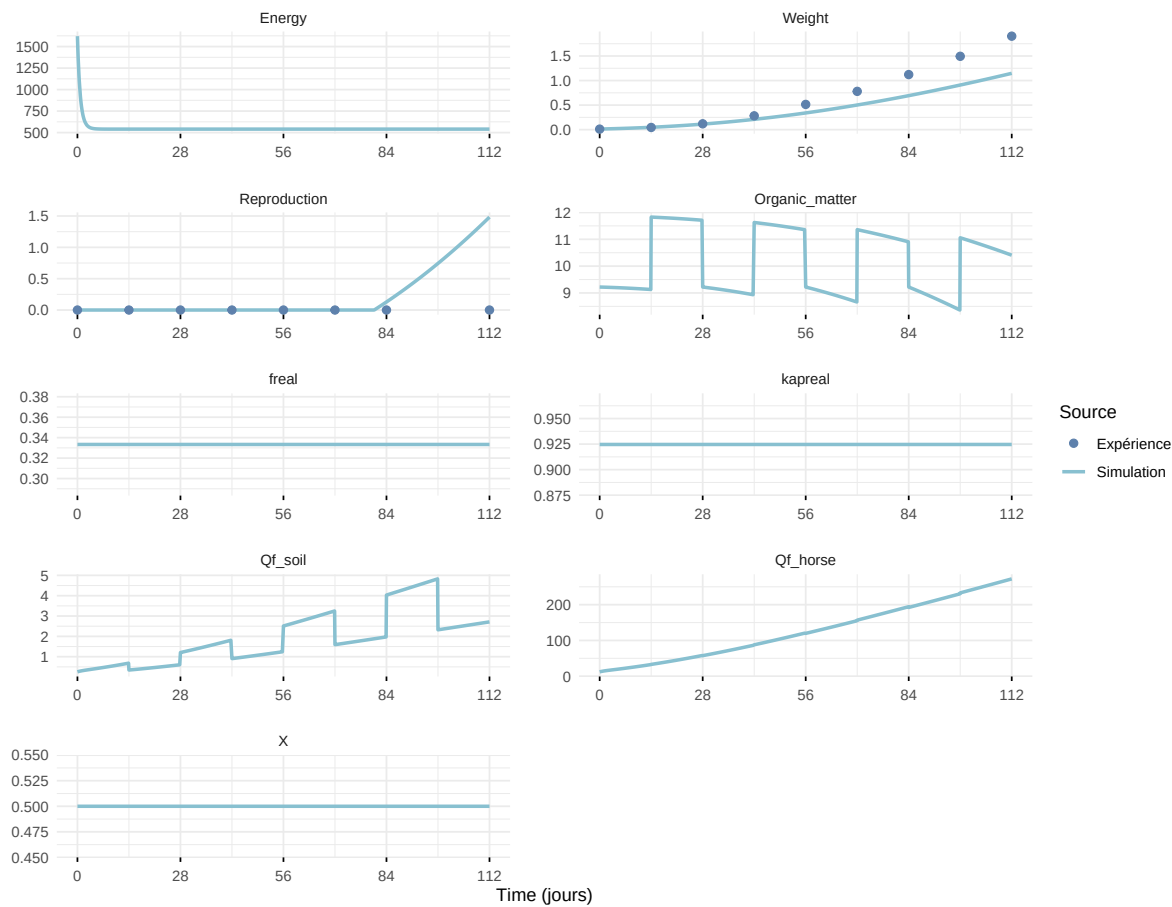
Expérience : Bart2019_XP5



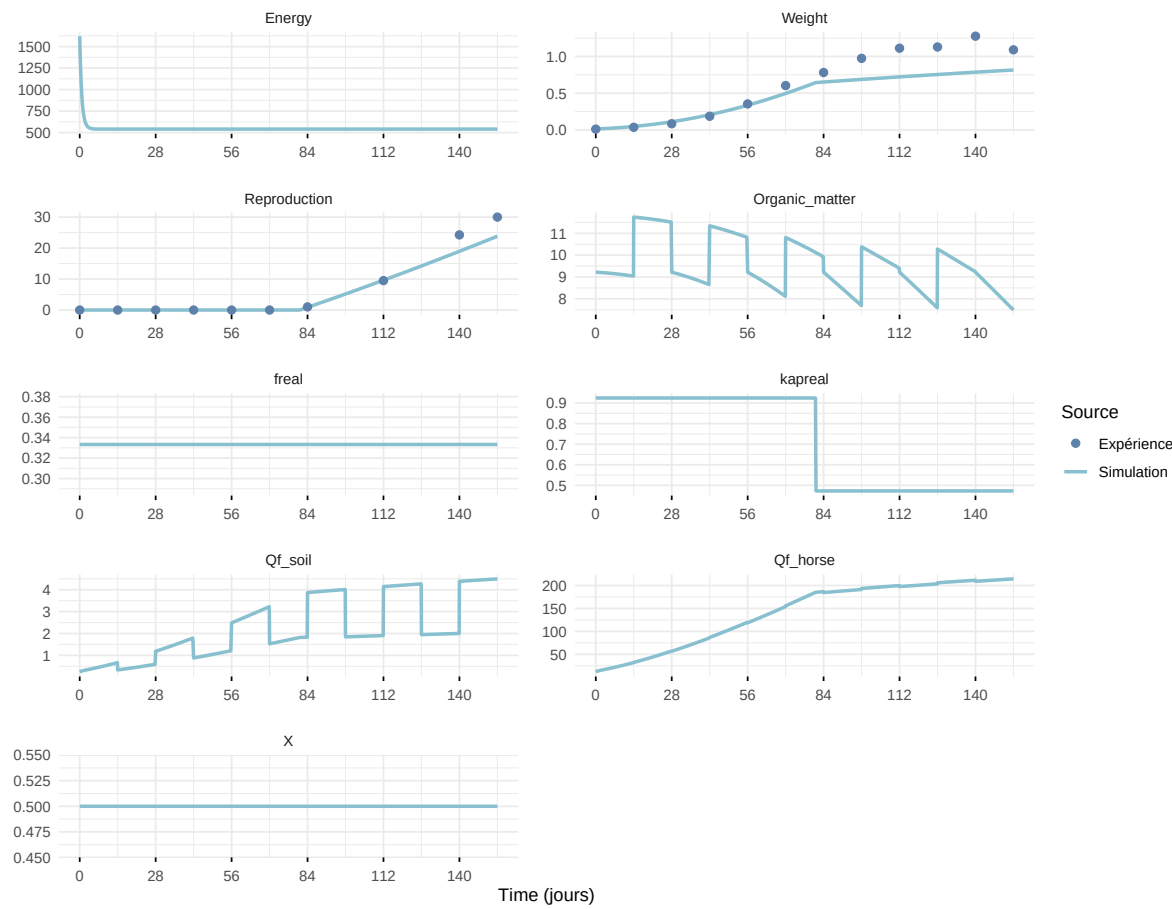
Expérience : Bart2020



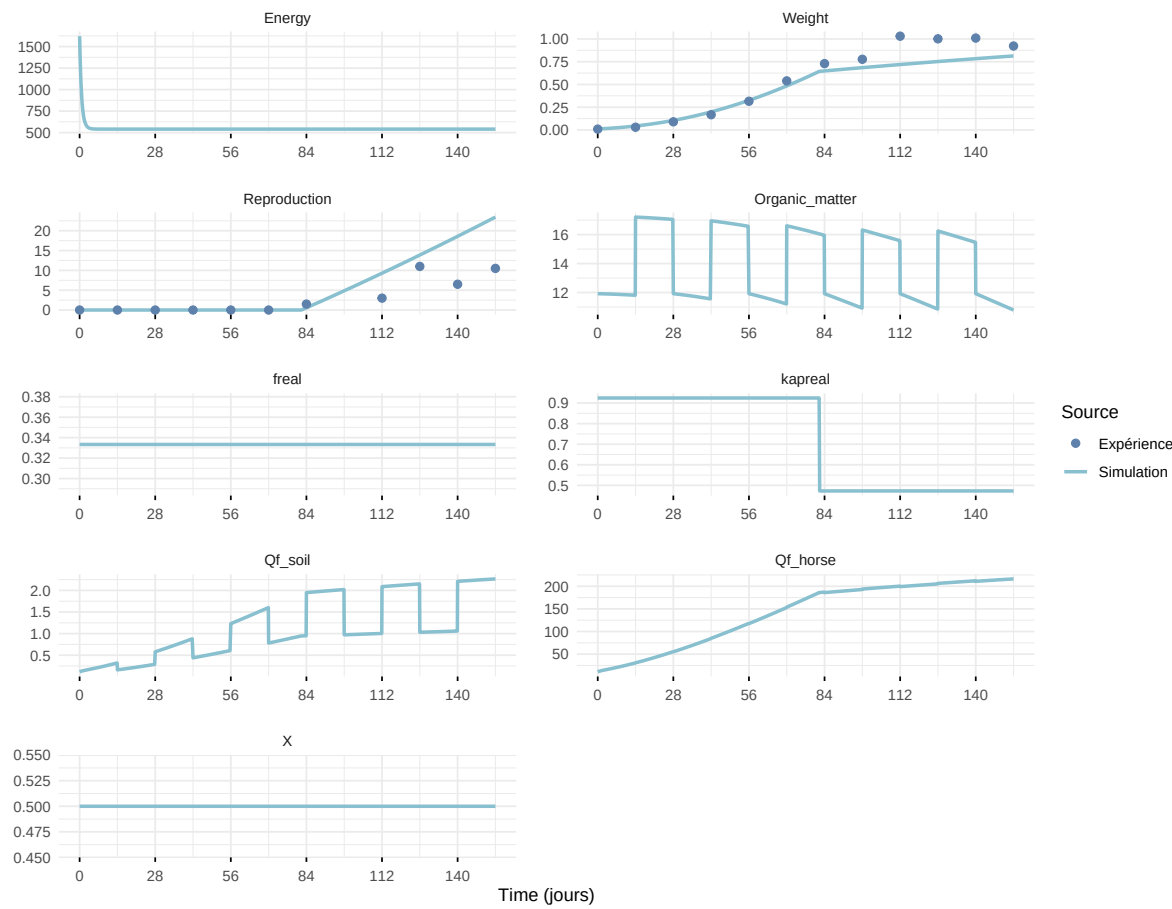
Expérience : Gollot2026_dens_D1



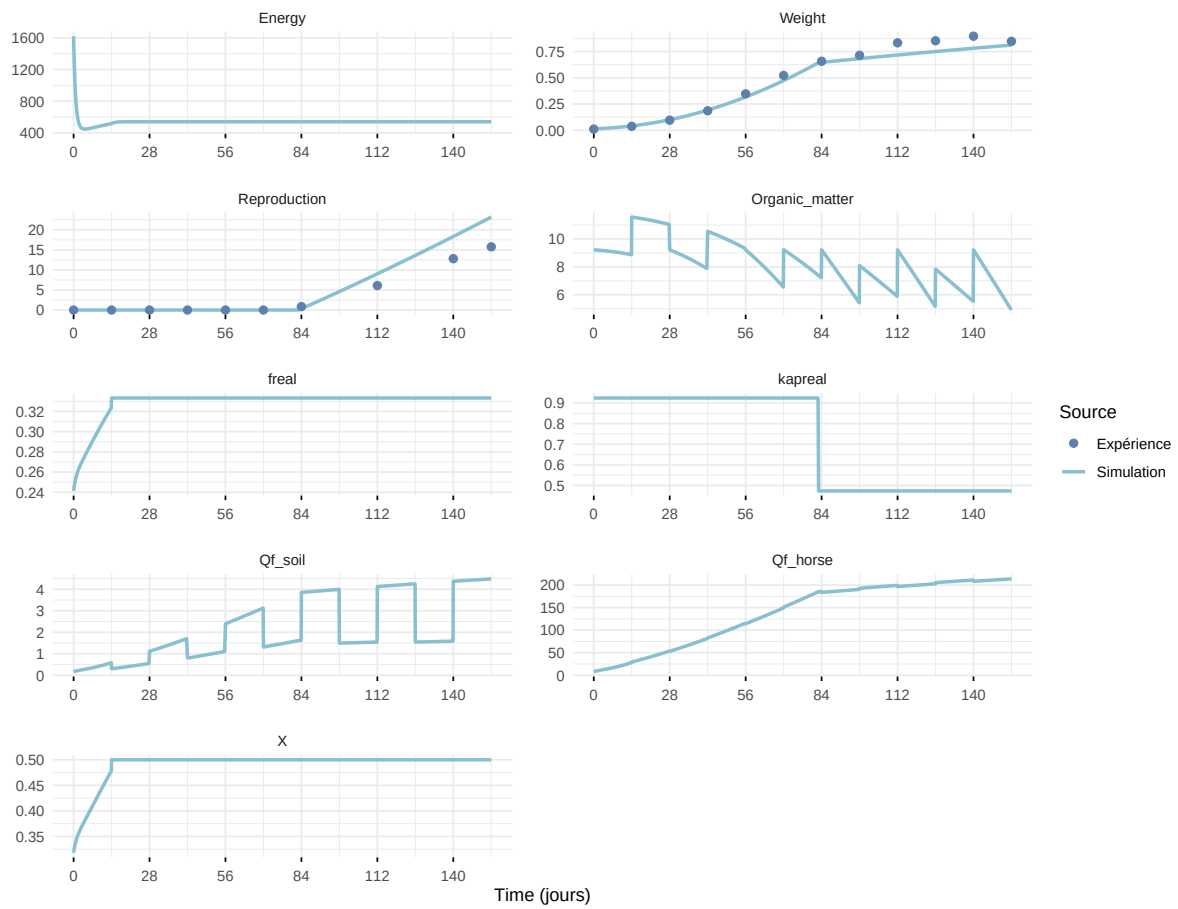
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

